

Online observations

RGB stream



K1 I2 K3 I4 K5

Arrived views

K keyframe **I** intermediate

Online
frontend

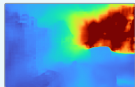
Estimated poses



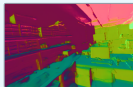
Start

End

Keyframe depth



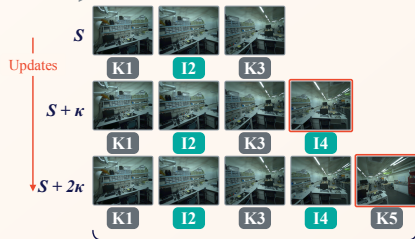
Keyframe normal



Training-view management

(a) View Growth

Retain old views; admit arrived views



(b) View Sampling



Gaussian map optimization

Completed updates

Shared Gaussian map



C07 region · cutaway visualization

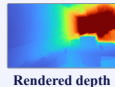
Update

Selected camera

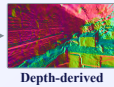
Render



Rendered RGB



Rendered depth



Depth-derived normal

Base supervision

RGB (K + I) · Depth / normal (K)

Map objective +

Priors

(c) Ray-space geometry supervision

